

BankCore Digital Transformation Platform

IT Project Plan & Risk Analysis | Version 2.4

Project ID	BCDT-2025-001
Project Manager	Arjun Mehta
Sponsor	CTO Office, BankCore Systems Ltd
Classification	CONFIDENTIAL
Version	2.4 Final Approved
Prepared by	Program Management Office (PMO)
Report Date	August 15, 2025
Project Start	March 1, 2025
Planned Go-Live	November 30, 2025
Total Budget	USD 12,500,000

Current Risk Level: HIGH

1. Executive Summary

The BankCore Digital Transformation Platform (BCDT) is a USD 12.5M, 9-month initiative to modernise BankCore Systems Ltd's core banking infrastructure. The project replaces a 27-year-old COBOL mainframe with a cloud-native microservices architecture on AWS, integrating real-time payments (ISO 20022), open banking APIs (PSD2), and an AI-powered fraud detection engine.

As of August 2025 the project is 62% complete. Four of eleven workstreams are behind schedule. The Regulatory Compliance and Data Migration workstreams present the highest risk. Budget utilisation stands at 71% (USD 8.9M spent), with a projected 8% overrun if current velocity is not improved.

Key Project Vitals

Overall Progress	62%
Current Phase	Phase 3 -- Core Migration & Integration
Schedule Variance	-14 days (behind schedule)
Budget Utilisation	71% (USD 8.9M / 12.5M)
Projected Overrun	USD 950,000 (+8%)
Open Issues	7 (2 Critical, 3 High, 2 Medium)
Team Size	68 FTEs + 12 external consultants
Key Stakeholders	CTO, CFO, RBI Compliance Officer, AWS Architecture Team

2. Project Scope & Objectives

2.1 In-Scope

- Decommission legacy IBM z14 mainframe and migrate 14.2M customer accounts to cloud-native core.
- Build and deploy 23 microservices: accounts, loans, payments, FX, notifications, identity, fraud.
- Implement ISO 20022 real-time payment rails with SWIFT connectivity.
- Develop open banking API gateway (PSD2/RTS) for third-party fintech integrations.
- AI-powered fraud detection engine using XGBoost -- real-time scoring under 50ms latency.
- Customer-facing mobile and web banking portal redesign (React Native / Next.js).
- Staff training for 1,400 branch employees and 220 IT personnel.
- RBI and SEBI compliance certification including data localisation mandates.
- Disaster recovery: RPO < 4 hours, RTO < 2 hours, 99.99% SLA on core banking.

2.2 Out-of-Scope

- Physical branch infrastructure upgrades (separate Facilities project).
- ATM network software (separate vendor contract -- NCR).
- International subsidiary systems (Phase 2, FY2026).
- Insurance or brokerage platform integrations.

2.3 Key Deliverables

- D1: Cloud infrastructure setup (AWS VPC, Transit Gateway, Direct Connect) -- COMPLETED
- D2: Core banking microservices (Accounts, Loans, FX) -- IN PROGRESS (72%)
- D3: Payment processing engine (NEFT, RTGS, UPI, ISO 20022) -- IN PROGRESS (58%)
- D4: Fraud detection AI model -- COMPLETED
- D5: Open banking API gateway -- IN PROGRESS (44%)
- D6: Mobile and web portal -- IN PROGRESS (61%)
- D7: Data migration: 14.2M accounts from mainframe to PostgreSQL/Aurora -- IN PROGRESS (38%)
- D8: RBI Compliance certification -- DELAYED (25%, 21-day lag)
- D9: Performance and penetration testing -- NOT STARTED
- D10: Staff training and change management -- IN PROGRESS (50%)
- D11: Go-live cutover and hypercare -- NOT STARTED

3. Project Schedule & Milestones

The project follows a phased delivery model. A 21-day schedule variance was recorded in June 2025 due to delays in RBI regulatory sign-off and data quality issues in the mainframe-to-cloud migration pipeline.

ID	Milestone	Start	Plan End	Forecast	Status	Var.
M1	Project Kickoff & Cloud Infra	Mar 1, 2025	Mar 28	Mar 26	COMPLETED	+2d
M2	Phase 1: Infra & DevOps Pipeline	Apr 1, 2025	Apr 30	Apr 30	COMPLETED	On time
M3	Core Banking Microsvcs (Alpha)	May 1, 2025	Jun 15	Jun 19	COMPLETED	+4d
M4	Fraud AI Model Deployment	Jun 1, 2025	Jun 30	Jul 2	COMPLETED	+2d
M5	Payment Engine Integration	Jul 1, 2025	Aug 15	Sep 5	DELAYED	+21d
M6	Data Migration -- 50% accounts	Jul 15, 2025	Sep 1	Sep 22	DELAYED	+21d
M7	Open Banking API Gateway (Beta)	Aug 1, 2025	Sep 30	Oct 14	AT RISK	+14d est
M8	RBI Compliance Certification	Sep 1, 2025	Oct 15	Nov 5	AT RISK	+21d est
M9	Pen & Performance Testing	Oct 1, 2025	Nov 1	TBD	NOT STARTED	-
M10	User Acceptance Testing	Oct 15, 2025	Nov 15	TBD	NOT STARTED	-
M11	Production Go-Live	Nov 30, 2025	Nov 30	TBD	AT RISK	+14-21d est

4. Team Structure & Resource Allocation

The project is organised into 11 workstreams. External consultants from AWS Professional Services and Deloitte Cybersecurity are embedded in specific workstreams. Total team: 68 FTEs + 12 contractors.

Workstream	Lead	FTEs	Avail.	Status	Progress
Core Banking Engineering	Ravi Kumar	18	80%	IN PROGRESS	72%
Payment Systems	Priya Sharma	12	70%	DELAYED	58%
Fraud AI & Data Science	Deepak Nair	8	100%	COMPLETED	100%
Open Banking API	Sneha Patel	7	65%	IN PROGRESS	44%
Mobile & Web Frontend	Aditya Roy	10	85%	IN PROGRESS	61%
Data Migration	Kiran Rao	9	60%	DELAYED	38%
Cloud Infrastructure	AWS Pro Svcs	5	100%	COMPLETED	100%
Security & Pen Testing	Deloitte Cyber	6	40%	NOT STARTED	0%
RBI Compliance & Legal	Meera Joshi	4	50%	DELAYED	25%
QA & Performance Testing	Suresh Menon	6	40%	NOT STARTED	0%
Training & Change Mgmt	Anita Gupta	5	75%	IN PROGRESS	50%

Critical resource gap: The Data Migration team is at 60% capacity due to two senior DBAs on medical leave. Augmentation request for 3 contract DBAs pending CTO approval (Issue BCDT-ISS-019).

5. Risk Register

Likelihood 1-5 (1=Rare, 5=Almost Certain). Impact 1-5 (1=Negligible, 5=Catastrophic). Risk Score = Likelihood x Impact.

ID	Risk Description	Category	L	I	Score	Level	Mitigation
R01	RBI compliance cert. delayed beyond C	Regulatory	5	5	25	CRITICAL	Weekly RBI checkpoints; dedicated regulatory liaison.
R02	Data corruption during mainframe cuto	Technical	3	5	15	HIGH	Dual-write strategy; automated reconciliation; rollback p
R03	Payment engine latency >200ms under	Performance	4	4	16	HIGH	Load test Oct 1; Redis caching layer under evaluation.
R04	Key personnel attrition -- Data Migration	Resource	3	4	12	HIGH	Retention bonus approved; knowledge transfer mandate
R05	Open Banking API security vulnerability	Security	3	4	12	HIGH	Pen test by Deloitte Sep 15; bug bounty programme.
R06	Third-party fintech partner integration d	External	4	3	12	MEDIUM	SLA penalties in contracts; sandbox provided to partner
R07	AWS region outage during migration wi	Infrastructure	2	5	10	MEDIUM	Multi-AZ; DR in ap-south-2; tested quarterly.
R08	Budget overrun >15%	Financial	3	3	9	MEDIUM	Weekly budget review; change control board for scope
R09	Staff resistance during UAT	Change Mgmt	3	3	9	MEDIUM	Change champion network; hands-on training; feedback
R10	Fraud model drift post go-live	AI/ML	2	3	6	LOW	Automated monitoring; retrain trigger if F1 < 0.90.

6. Budget Breakdown & Spend Analysis

Workstream / Category	Approved (USD)	Spent (USD)	Remaining (USD)	% Spent
Cloud Infrastructure (AWS)	2,200,000	1,980,000	220,000	90%
Core Banking Engineering	3,500,000	2,520,000	980,000	72%
Payment Systems Development	1,800,000	1,044,000	756,000	58%
Fraud AI & Data Science	600,000	600,000	0	100%
Open Banking API	750,000	330,000	420,000	44%
Mobile & Web Frontend	800,000	488,000	312,000	61%
Data Migration	900,000	342,000	558,000	38%
Security Testing	400,000	60,000	340,000	15%
RBI Compliance & Legal	350,000	87,500	262,500	25%
Training & Change Mgmt	300,000	150,000	150,000	50%
PMO & Governance	400,000	280,000	120,000	70%
Contingency Reserve (10%)	500,000	18,000	482,000	4%
TOTAL	12,500,000	7,899,500	4,600,500	63%

Projected overrun: USD 950,000 (+7.6%). Drivers: payment engine rework (USD 380K), extended RBI compliance engagement (USD 220K), data migration tooling upgrade (USD 190K). Change request CR-2025-014 pending CFO approval.

7. Technical Architecture Overview

7.1 Cloud Platform

Primary cloud: AWS ap-south-1 (Mumbai). DR: AWS ap-south-2 (Hyderabad). Hub-and-spoke VPC topology with AWS Transit Gateway connecting 6 VPCs: Core Banking, Payments, Fraud, API Gateway, Data Lake, and Management/DevOps.

7.2 Microservices Stack

- Runtime: Java 21 (Spring Boot 3.2) for core banking; Python 3.11 (FastAPI) for AI/ML services.
- Container orchestration: Amazon EKS (Kubernetes 1.30) with Karpenter autoscaling.
- Service mesh: AWS App Mesh (Envoy proxy) for east-west traffic with mTLS.
- API Gateway: Kong Enterprise for PSD2 open banking; AWS API Gateway for internal services.
- Database: Amazon Aurora PostgreSQL 15 (multi-region) for transactions; DynamoDB for sessions; Redis 7 for caching.
- Message broker: Apache Kafka on MSK (3-broker cluster) for event streaming.
- Observability: Datadog APM + AWS CloudWatch + PagerDuty for on-call alerting.

7.3 AI Fr

Two-stage approach: (1) Real-time XGBoost classifier scoring each transaction in under 50ms using 47 engineered features (velocity, geolocation, device fingerprint, historical behaviour). (2) Offline ensemble model (XGBoost + Isolation Forest) for nightly batch fraud sweeps on the full transaction graph. Results: 94.2% precision, 91.8% recall on holdout test set (2.1M synthetic + 800K historical transactions). Automated retraining triggers if F1 drops below 0.90.

7.4 Data Migration Strategy

Strangler-fig migration across 6 batches (~2.3M accounts each). Each batch: extract (IBM DataStage) -> transform (dbt on Redshift) -> validate (140 data quality rules) -> load (Aurora PostgreSQL). Dual-write proxy ensures zero data loss during migration window. Full cutover planned for a 72-hour maintenance window in November 2025.

8. Open Issues & Action Items

Issue ID	Description	Sev.	Owner	Due	Status / Action
ISS-012	Payment engine throughput drop under 10K TP	CRITICAL	Ravi Kumar	Aug 20	Redis caching in progress; load test Aug
ISS-019	Data migration team resource gap (2 DBAs on	CRITICAL	Kiran Rao	Aug 18	Augmentation request to CTO; contractor
ISS-023	RBI compliance SLA breach (14 days overdue)	HIGH	Meera Joshi	Aug 25	Escalation letter sent; CTO meeting RBI
ISS-027	ISO 20022 FX message validation failures	HIGH	Priya Sharma	Aug 28	SWIFT team engaged; schema mapping
ISS-031	Open Banking sandbox SSL certificate expiry	HIGH	Sneha Patel	Aug 18	Renewal raised with DevOps; resolution
ISS-034	Mobile app iOS 18 compatibility regression	MEDIUM	Aditya Roy	Sep 1	iOS 18 beta added to QA pipeline; fix in S
ISS-037	Training content not localised for Tamil Nadu b	MEDIUM	Anita Gupta	Sep 5	Tamil translation vendor engaged; review

9. Recommendations & Next Steps

9.1 Immediate Actions (Next 2 Weeks)

- Resolve DBA gap by engaging 2 contract DBAs from TechM by Aug 22. Owner: Kiran Rao
- Emergency load test of payment engine at 12K TPS by Aug 22. Owner: Ravi Kumar
- Escalate RBI compliance delay to Board level; request expedited review. Owner: CTO
- Fix ISO 20022 FX schema mapping and rerun end-to-end payment tests. Owner: Priya Sharma
- Renew Open Banking sandbox SSL certificates immediately. Owner: DevOps Lead

9.2 Sche

A schedule recovery task force has been formed. Recommended approach: (a) parallel execution of Data Migration Batch 3 and API Gateway beta rollout; (b) 10% budget reallocation from contingency to augment payment and migration teams; (c) scope reduction of Open Banking API to 8 certified partners at go-live (from 14), with remaining partners onboarded in Phase 2 (Q1 2026).

9.3 Go/No-Go Criteria for November 30 Go-Live

- RBI compliance certification received by October 31, 2025.
- All 14.2M accounts migrated and reconciled with zero critical data quality failures.
- Payment engine sustains 12,000 TPS at p99 latency < 200ms over 30-minute sustained load.
- Fraud detection model live with F1 score >= 0.91 on production traffic.
- Penetration test completed with zero Critical or High severity open findings.
- UAT sign-off obtained from all 5 business unit heads.
- Disaster recovery drill completed with RTO < 2 hours validated.

9.4 Cont

If go-live criteria cannot be met by Nov 30: Phase A -- Retail accounts (8M) go live Dec 15, 2025; Phase B -- Corporate and SME accounts (5M) go live Jan 31, 2026; Phase C -- International correspondent banking Q2 2026. Pre-approved by Board Risk Committee.

10. Approvals & Sign-Off

Name	Role	Decision	Date
Arjun Mehta	Project Manager	Approved	Aug 15, 2025
Rajesh Sharma	CTO	Approved	Aug 15, 2025
Kavitha Nair	CFO	Conditional -- CR-2025-014	Aug 15, 2025
Dr. Pradeep Iyer	Board Risk Committee Chair	Reviewed	Aug 14, 2025

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